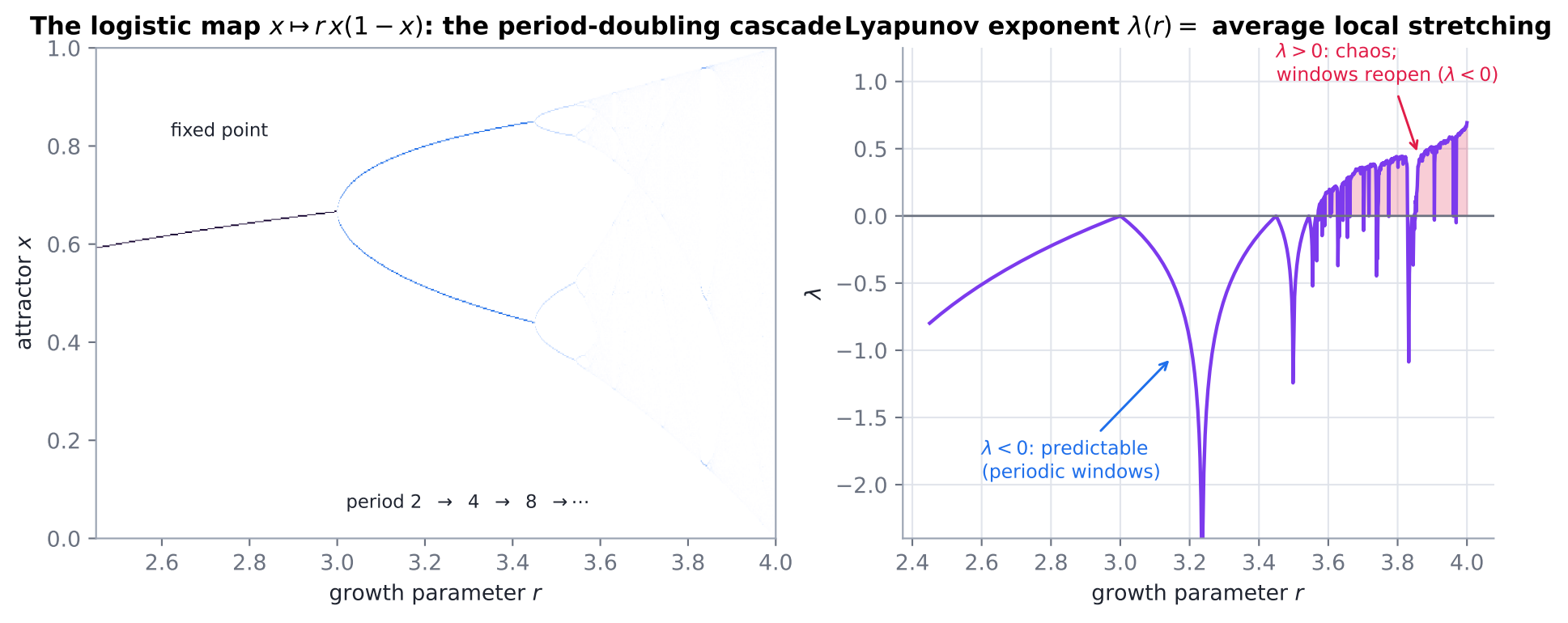

The logistic map $x \mapsto r\,x(1-x)$: the period-doubling cascade
fixed point
period 2 → 4 → 8 →⋯
attractor $x$
growth parameter $r$
Lyapunov exponent $\lambda(r) =$ average local stretching
$\lambda > 0$: chaos;
windows reopen ($\lambda < 0$)
$\lambda < 0$: predictable
(periodic windows)
$\lambda$
growth parameter $r$